

Literature Resolution of Remark 7.3

Right Bregman Klee Sets without Full Domain

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Status. `literature_already_answered`.

Original question

Bauschke, Wang, Ye, and Yuan assume that f is a Legendre function on $\mathbb{R}^J$, write $U = \text{int dom } f$, and take a nonempty compact set $C \subset U$. Their Theorem 7.2 proves that a right Bregman Klee set is a singleton when $\text{dom } f = \mathbb{R}^J$. Remark 7.3 on PDF page 15 then asks whether the theorem remains true when the full-domain assumption is dropped [1].

Here “right Bregman Klee” means that, for every $x \in U$, the maximizer of

$$c \mapsto D_f(x, c) \quad (c \in C)$$

is unique.

Explicit supporting answer

Bauschke, Macklem, Sewell, and Wang explicitly identify “[7, Remark 7.3]” and state on PDF page 4 that they settle the question affirmatively without full domain. Their Theorem 3.2 on PDF page 10 says [2]:

Suppose that C is compact and that C is right Bregman Klee. Then C is a singleton.

This has exactly the hypotheses left after removing $\text{dom } f = \mathbb{R}^J$ from the source theorem, so it is a direct and complete answer.

Proof mechanism

The later paper considers the convex farthest-distance function

$$F_C(x) = \sup_{c \in C} D_f(x, c).$$

Its Theorem 3.1 proves that F_C has a minimizer x_0 in U , even when $\text{dom } f$ has a boundary. The proof uses coercivity of each $D_f(\cdot, c)$ and a Bregman projection argument to rule out a boundary minimizer. At x_0 , the Ioffe–Tikhomirov subdifferential formula and the unique active farthest point $Q_C(x_0)$ give

$$0 \in \partial F_C(x_0) = \{\nabla f(x_0) - \nabla f(Q_C(x_0))\}.$$

The Legendre gradient is injective, hence $x_0 = Q_C(x_0)$. Thus the largest value of $D_f(x_0, c)$ over C is $D_f(x_0, x_0) = 0$; nonnegativity and strict convexity force $C = \{x_0\}$.

Scope

The answer covers general finite-dimensional Legendre functions in the later paper’s standing setting, including non-full-domain examples such as Kullback–Leibler and Itakura–Saito divergences. It does not settle the classical infinite-dimensional Hilbert-space farthest-point conjecture.

References

- [1] H. H. Bauschke, X. Wang, J. Ye, and X. Yuan, *Bregman distances and Klee sets*, arXiv:0802.2322.
- [2] H. H. Bauschke, M. S. Macklem, J. B. Sewell, and X. Wang, *Klee sets and Chebyshev centers for the right Bregman distance*, arXiv:0908.2013.